



Vendor: Microsoft

Exam Code: AI-300

Exam Name: Machine Learning Operations (MLOps)

Engineer Associate

Version: DEMO

QUESTION 1

Case Study 1 - Fabrikam Inc.

Background

Fabrikam Inc. is a mid-sized healthcare analytics company that provides population health dashboards and predictive insights to regional hospital systems across the United States. Fabrikam Inc. customers rely on near real time analytics to monitor patient flow, staffing needs, and readmission risks. They use multiple traditional forecasting machine learning models for predictions.

Fabrikam Inc. has an established Microsoft Azure footprint. The company uses Jupyter Notebooks that run on a local server as the primary development environment. The data science team is experiencing scalability, asset management and code management issues with the current development platform. Fabrikam Inc. plans to migrate to a cloud-based development environment to mitigate the issues.

Additionally, the company plans to implement a Retrieval-Augmented Generation (RAG)-based chat application for client support. Leadership requires the application to be developed and deployed with a low operational risk.

Current Environment

Fabrikam Inc. operates a single Azure subscription that has the following components:

- Azure Data Lake Storage Gen2 that contains de-identified clinical and operational datasets
- Azure AI Search indexing curated analytical documents and reference materials
- A small set of Python-based training scripts maintained by data scientists
- Azure OpenAI Service with deployed foundational models
- A Microsoft Foundry resource for building a RAG-based solution

Evaluation data has manually defined expected responses.

The current challenges faced by the data science team include the following:

- Model training jobs are run manually from notebooks.
- Experiment tracking is inconsistent
- Model versions are registered without standardized metadata.
- Deployment is performed manually by data scientists, with limited rollback capability.
- The team has no standardized evaluation process for generative AI outputs.

The environment currently allows public network access. Authentication relies on user accounts rather than managed identities. Compute targets are manually created and shared across experiments. This has led to resource contention during peak usage.

Business Requirements

Fabrikam Inc. has the following business requirements for the modernization initiative:

- Provide a conversational interface that answers analytics questions by using internal documents and datasets.
- Ensure that sensitive healthcare-related data is not exposed outside the Fabrikam Inc. Azure tenant.
- Enable repeatable and auditable model training and deployment processes.
- Support experimentation to compare prompt strategies and fine-tuned models.
- Align the model with the ranked preferences and optimize behavior for the long term.

- Minimize disruption to existing analytics workloads during rollout.

Technical Requirements

To support the business goals, Fabrikam Inc. identifies these technical requirements:

- Use Azure Machine Learning workspaces to centrally manage data assets, models, and environments.
- Implement experiment tracking and model versioning for all training jobs.
- Orchestrate training and evaluation by using pipelines rather than manually running notebooks.
- Deploy traditional machine learning models with support for staged rollout and rollback.
- Improve RAG-based solution output quality.
- Use the existing evaluation datasets that are based on real data with input-output pairs.
- Apply advanced fine-tuning techniques only when prompt engineering is insufficient

Issues and Constraints

Fabrikam Inc. must comply with internal security policies that require the company to restrict network access and avoid long-lived secrets. The data science team has limited Azure DevOps experience, so solutions must favor managed services and automation over custom infrastructure.

Cost predictability is important. Leadership prefers serverless or managed compute options where possible but is willing to approve dedicated compute for stable production workloads.

Problem Statement

Fabrikam Inc. must design and implement an Azure-based AI operations solution that enables reliable training, evaluation, deployment, and iteration of generative AI models. The solution must support experimentation and gradual rollout while ensuring governance, security, and operational stability. The data science and platform teams must collaborate to deliver this solution by using Azure Machine Learning and Microsoft Foundry capabilities.

You need to standardize how Fabrikam Inc. manages machine learning assets. Which action should you perform first?

- A. Register assets in the Azure Machine Learning registry.
- B. Create a shared Azure Machine Learning workspace.
- C. Deploy a managed online endpoint.
- D. Create a new Microsoft Foundry project.

Answer: B

Explanation:

Scenario: To support the business goals, Fabrikam Inc. identifies these technical requirements: Use Azure Machine Learning workspaces to centrally manage data assets, models, and environments.

To centrally manage data assets, models, and environments across multiple Azure Machine Learning workspaces, you should Create a shared Azure Machine Learning workspace first.

The workspace serves as the top-level resource for your machine learning activities, providing a centralized place to view and manage the artifacts you create. While Registries are used to share assets (like models and environments) across existing workspaces, you must have a workspace as a prerequisite to create or use those assets in a project context.

Key Management Options

Azure provides several ways to organize and centralize your machine learning operations:

Shared Workspace: The primary container for managing data, compute, and experiments within a project team.

Registries: Used specifically for MLOps to decouple assets from specific workspaces, allowing them to be promoted through development, test, and production environments.

Hub Workspaces: A newer feature that groups multiple project workspaces under a single "hub" to share security settings, connections, and compute resources.

Reference:

<https://docs.azure.cn/en-us/machine-learning/concept-workspace>

QUESTION 2

You train models on GPU-enabled clusters but deploy them on CPU-based endpoints. Recently, inference failures occur due to incompatible dependencies. What should you do to ensure consistency?

- A. Use same compute for training and inference
- B. Define and reuse environment configurations
- C. Increase endpoint compute size
- D. Use batch endpoints

Answer: B

Explanation:

Defining and reusing environment configurations ensures that dependencies remain consistent between training and inference. This prevents runtime errors caused by mismatched libraries. Using identical compute resources is unnecessary and inefficient, as consistency depends on environment configuration rather than hardware.

QUESTION 3

Hotspot Question

A team trains an MLflow model that scores customer churn risk. The model will be consumed by different downstream systems.

One system requests predictions synchronously during customer interactions.

Another system submits files containing millions of records for scheduled scoring.

You need to deploy the model by using managed inference options that match each usage pattern.

Which option should you use for each usage pattern? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Managed inference options

Requirement

Low-latency synchronous predictions

Deployment option

Batch endpoint

Job-based training pipeline

Real-time endpoint

Registered model artifact

High-volume scheduled scoring

Batch endpoint

Online endpoint with autoscaling

Managed compute cluster

Model registry version

Answer:

Managed inference options

Requirement

Low-latency synchronous predictions

Deployment option

Batch endpoint

Job-based training pipeline

Real-time endpoint

Registered model artifact

High-volume scheduled scoring

Batch endpoint

Online endpoint with autoscaling

Managed compute cluster

Model registry version

QUESTION 4

You manage an Azure Machine learning workspace. You develop a machine learning model.

You must deploy the model to use a low-priority VM with a pricing discount.

You need to deploy the model.

Which compute target should you use?

- A. Azure Container Instances (ACI)
- B. Azure Machine Learning compute clusters
- C. Local deployment
- D. Azure Kubernetes Service (AKS)

Answer: B

Explanation:

The best compute target for deploying a model using low-priority VMs (or their modern successor, Spot VMs) is an Azure Machine Learning compute cluster.

Best Compute Target: AML Compute Cluster

For low-priority/Spot pricing, you should use an Azure Machine Learning compute cluster configured with the LowPriority tier.

Primary Use Case: This target is specifically recommended for batch deployments. Batch inference is ideal for low-priority VMs because these jobs are asynchronous and can tolerate the interruptions (preemptions) inherent to discounted capacity.

Pricing Advantage: Low-priority VMs offer significant discounts—often up to 80% off standard rates—by utilizing unused Azure capacity.

Automatic Handling: When a node is preempted during a batch job, Azure Machine Learning automatically attempts to replace the lost capacity and re-queues failed tasks to the cluster.

Reference:

<https://learn.microsoft.com/en-us/azure/machine-learning/how-to-use-low-priority-batch>

QUESTION 5

A team manages an Azure Machine Learning workspace where they deploy models to online endpoints.

The team needs to introduce a new version of a model to production without disrupting existing users.

The team must validate the new version before full rollout.

You need to reduce risk during deployment.

What should you do?

- A. Deploy the model to a batch endpoint.
- B. Split traffic between deployments.
- C. Replace the existing endpoint.
- D. Route all traffic to the new deployment.

Answer: B

Explanation:

To introduce a new model version in Azure Machine Learning without service interruption, you should use Blue/Green Deployment with Traffic Splitting.

This strategy allows you to run two versions of a model simultaneously under a single Online Endpoint, gradually shifting users to the new version once it is validated.

Key Benefits

Zero Downtime: The endpoint URL stays the same; only the backend routing changes.

Easy Rollback: If the new model fails, you can instantly flip traffic back to 100% on the old version.

Risk Mitigation: Only a small subset of users is exposed to the unproven model initially.

Implementation Steps

1. Create the "Green" Deployment

Deploy the new model version as a second deployment under the existing online endpoint. Initially, set its traffic allocation to 0%.

2. Canary Testing (Initial Split)

Shift a small percentage of traffic (e.g., 10%) to the new deployment. Monitor performance metrics, error rates, and model accuracy in a real-world environment.

3. Validation & Monitoring

Use Azure Monitor and Application Insights to compare the two deployments. Check for:

Latency: Is the new model slower?

HTTP Status Codes: Are there 4xx or 5xx errors?

Model Drift: Is the prediction quality as expected?

4. Full Rollout

If the new version is stable, increase the traffic split (e.g., 50/50) until the new model handles 100% of the traffic.

5. Cleanup

Once the "Green" deployment is confirmed as the new production standard, you can delete the old ("Blue") deployment to save costs.

Reference:

<https://learn.microsoft.com/en-us/azure/well-architected/ai/operations>

QUESTION 6

You have a deployment of an Azure OpenAI Service base model.

You plan to fine-tune the model.

You need to prepare a file that contains training data.

Which file format should you use?

- A. CSV
- B. TSV
- C. JSONL
- D. JSON

Answer: C

Explanation:

To fine-tune a model in the Azure OpenAI Service, your training data must be in JSONL (JSON

Lines) format.

File Requirements

Format: Each line of the file must be a single, valid JSON object.

Encoding: The file must be encoded in UTF-8.

Special Requirement: For some older tools or specific Azure OpenAI use cases, the file may need to include a byte-order mark (BOM).

Size Limits: Individual files typically have a maximum size (e.g., 100 MB per file in some contexts), but you can often upload multiple files.

Reference:

<https://dev.to/icebeam7/fine-tuning-a-model-with-azure-open-ai-studio-39p7>

QUESTION 7

You have a deployment of an Azure OpenAI Service base model.

You plan to fine-tune the model.

You need to prepare a file that contains training data for multi-turn chat.

Which file encoding method should you use?

- A. ISO-8859-1
- B. UTF-16
- C. UTF-8
- D. ASCII

Answer: C

Explanation:

For preparing a multi-turn training data file for the Azure OpenAI Service, you should use UTF-8 with a Byte Order Mark (BOM) encoding.

File Format Requirements

Format: The file must be in JSON Lines (JSONL) format, where each individual line is a valid JSON object representing one training example.

Encoding: Specifically, Azure OpenAI requires the JSONL file to be encoded in UTF-8 with BOM.

Structure: For multi-turn conversations, each line must contain a messages array with multiple role ("system", "user", "assistant") and content pairs to represent the dialogue history.

Reference:

<https://dev.to/icebeam7/fine-tuning-a-model-with-azure-open-ai-studio-39p7>

QUESTION 8

You are fine-tuning a base language model to analyze customer feedback.

You label examples of support tickets. You must improve classification accuracy by configuring and fine-tuning the base model in Microsoft Foundry.

You need to configure and run fine-tuning.

What should you do first?

- A. Use prompt flow to generate multiple prompt templates for evaluation.
- B. Deploy the base model to an online endpoint before starting fine-tuning.
- C. Enable tracing for all inference calls in the evaluation pipeline.
- D. Format the dataset as a JSONL file with prompt-completion pairs and upload the file.

Answer: C

Explanation:

In Microsoft Foundry, when configuring and running a fine-tuning job for analyzing customer feedback (e.g., classifying support tickets), you should first enable tracing for all inference calls in the evaluation pipeline.

Tracing is a critical step in the "Evaluate" phase of the fine-tuning workflow, allowing you to capture input/output examples, identify the root cause of classification errors, monitor latency, and analyze model behavior before and after training.

Note:

To fine-tune a model and evaluate it effectively, you should follow this sequence:

1. Enable Tracing for Initial Baseline

Before you fine-tune, you should indeed enable tracing for all inference calls in your existing evaluation pipeline.

Purpose: This creates a clear "paper trail" of how the base model is currently failing.

Benefit: Tracing captures the exact inputs, outputs, and intermediate steps. By analyzing these traces, you can identify if the low accuracy is due to the model not following instructions, missing domain-specific jargon, or struggling with specific ticket categories.

Setup: You can enable Automatic Tracing (currently in preview) in Foundry to log these details to Application Insights without changing your code.

2. Configure and Run Fine-Tuning

Once you have analyzed the traces and prepared your labeled dataset, you can proceed with the fine-tuning job.

3. Continuous Evaluation

After the job completes, you must compare the fine-tuned model against your original traces

Reference:

<https://devblogs.microsoft.com/foundry/a-developers-guide-to-fine-tuning-gpt-4o-for-image-classification-on-azure-ai-foundry>

QUESTION 9

A team is working in Microsoft Foundry to test and compare large language model (LLM) prompt variants in a development environment.

The team requires consistent inputs to evaluate prompt variants without relying on live user traffic.

You need to create a controlled evaluation of input data.

Which action should you perform first?

- A. Generate synthetic interaction data.

- B. Configure content filters.
- C. Apply a blocklist.
- D. Enable observability metrics.

Answer: A

Explanation:

In Microsoft Foundry, you can test and compare LLM prompt variants using Prompt Flow. This environment allows you to create a controlled development workflow with consistent inputs, bypassing the need for live user traffic.

1. Generate Synthetic Interaction Data

Generating synthetic data is the recommended first step to build a robust, diverse test dataset without manual effort.

2. Compare Prompt Variants

Once your synthetic dataset is ready, use Prompt Flow to iterate on your prompts.

3. Controlled Evaluation

To measure performance objectively, run automated Eval Runs.

Reference:

<https://learn.microsoft.com/en-us/azure/foundry-classic/concepts/prompt-flow>

QUESTION 10

An organization maintains separate Azure Machine Learning workspaces for development and production.

Both environments must use the same validated assets without duplicating them.

Assets must be shared across workspaces while maintaining centralized governance and version control.

You need to enable reuse of assets across workspaces without copying them.

What should you do?

- A. Enable workspace-level Git integration and sync assets between repositories.
- B. Publish the asset as a pipeline component.
- C. Create a shared Azure Machine Learning environment that includes the asset.
- D. Publish the asset to an Azure Machine Learning registry.

Answer: D

Explanation:

To share validated assets between development and production workspaces while maintaining centralized governance and version control, you should use Azure Machine Learning Registries.

Unlike standard workspaces, registries are workspace-agnostic. They act as a central catalog that decouples assets from individual environments, allowing you to "promote" a model or environment from Dev to Prod without manual duplication or data drift.

Reference:

<https://learn.microsoft.com/en-us/azure/machine-learning/concept-machine-learning-registries-mlops>

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